RoboMaster Odometry - Phase 3

SCE-672 Advanced Robotics | Professor Aamer Iqbal Bhatti
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Objective and setup

The RoboMaster EP Core detected a fixed tag and drove straight toward it from the same **1.50 m** starting distance on eight successful trials. The robot stopped before the tag. Each post-stop estimated distance was compared with an independently taped distance from the **front bumper centre to the tag's flat front plane**.

The target was an upright AprilTag 36h11, ID 3, on a cardboard stand. Its measured black square was 0.142 m wide; its lower edge was about 0.01 m above the floor, giving an approximate centre height of 0.081 m. The tag, arm/camera pose, floor, and starting mark remained fixed. The nominal stop distance was 0.55 m, with a 0.08 m early-stop guard, so the robot stopped near an estimated 0.63 m. Forward speed was limited to 0.10 m/s; lateral and yaw commands were zero.

Camera measurement and calibration

OpenCV located ID 3 in 1280 x 720 frames and used its four corners, calibrated intrinsics, the measured square size, and IPPE square PnP to estimate camera-axis depth. The camera read the newest available frame. Two independently taped stationary placements mapped camera depth to bumper-to-tag distance:

d_camera = 0.97387735 x camera_axis_depth - 0.23056594 (metres).

The calibration pairs were 0.57 m tape / 0.8220 m depth and 1.50 m tape / 1.7770 m depth. Independent checks gave 0.3621 m against a 0.40 m tape reading (-0.0379 m) and 1.4776 m against 1.45 m (+0.0276 m). The conservative stop keeps the robot away from the closer range where the error grew. The older Phase 2 body transform failed a physical height check and was not used to calculate the forward-range correction; it remained in the detector as compatibility metadata and a forward-facing filter. Prior ToF readings were zero, so **ToF was not used** for correction.

Prediction, correction, and stopping

The SDK chassis-position subscription supplied forward odometry at 20 Hz. For odometry increment delta_x, the prediction was **d_pred = d_previous - delta_x**. A fresh camera range z corrected it as **d_new = d_pred + 0.35(z - d_pred)** if the innovation magnitude was at most 0.15 m. Otherwise the prediction was retained. Commands stopped on target guard, tag age over 0.50 s, odometry age over 0.30 s, or a 25 s time limit. A short post-stop sensing interval supplied the final estimate at the stopped pose.

Preflight

Stationary detection succeeded in 86/86 frames at 0.57 m, 113/113 at 1.50 m, and 111/111 at 0.40 m. A brief attended 0.06 m/s motion command stopped normally: tape displacement was about 0.05 m, while odometry reported +0.034 m. The operator reset the bumper to 1.50 m before each successful trial and taped the final distance before moving the robot again.

Eight successful trial results

Signed error = estimated final distance - measured final distance. Values below are rounded to 0.1 cm. The Trial 1 tape reading was reported as 'almost 61 cm'; the calculation uses 61 cm and is approximate.

Trial	Estimated (cm)	Measured (cm)	Error (cm)
1	63.2	~61.0	~+2.2
2	61.6	63.0	-1.4
3	61.9	65.0	-3.1
4	63.1	65.0	-1.9
5	61.6	64.0	-2.4
6	62.9	64.5	-1.6
8	62.8	64.5	-1.7
9	62.5	64.0	-1.5

Mean absolute error: 2.0 cm. **Signed error range:** -3.1 to +2.2 cm (width 5.3 cm). Statistics use the unrounded log values; the approximate first tape reading limits their precision.

Discussion and exception

All eight counted runs reached the target stop before the tag. The taped final distances ranged from 61.0 to 65.0 cm. Seven signed errors were negative, suggesting a small tendency to understate the actual stopping distance in this setup. Calibration/marker pose uncertainty, a slightly non-flat stand, tape placement, floor traction, odometry drift, frame timing, and motor stopping latency are plausible sources. The low-speed check showed 1.6 cm more movement by tape than by odometry. Camera correction limited prediction drift when fresh tag measurements arrived; prediction maintained the estimate between frames.

Attempt 7 stopped early at an estimated 0.832 m when a camera-frame pause exceeded the 0.50 s tag-age limit; the user measured about 0.85 m. It is retained as an aborted safety-stop attempt and excluded from the eight target-stop rows. An earlier Trial 2 connection attempt failed before motion when Windows switched away from RoboMaster Wi-Fi. No ToF measurement is claimed.

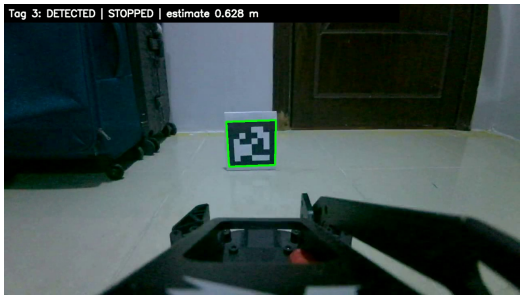
Robot-camera evidence

Trial 8 camera frames show the identified tag at the start and near the stop. The MP4 contains the full approach with ID 3 and estimated distance overlays.

Start



Near stop



Motion and distance tracking

The following graphs use Trial 8 raw telemetry. This representative run stopped at 0.628 m estimated versus 0.645 m measured. The target threshold is 0.63 m because it includes the 0.08 m guard added to the 0.55 m nominal stop.

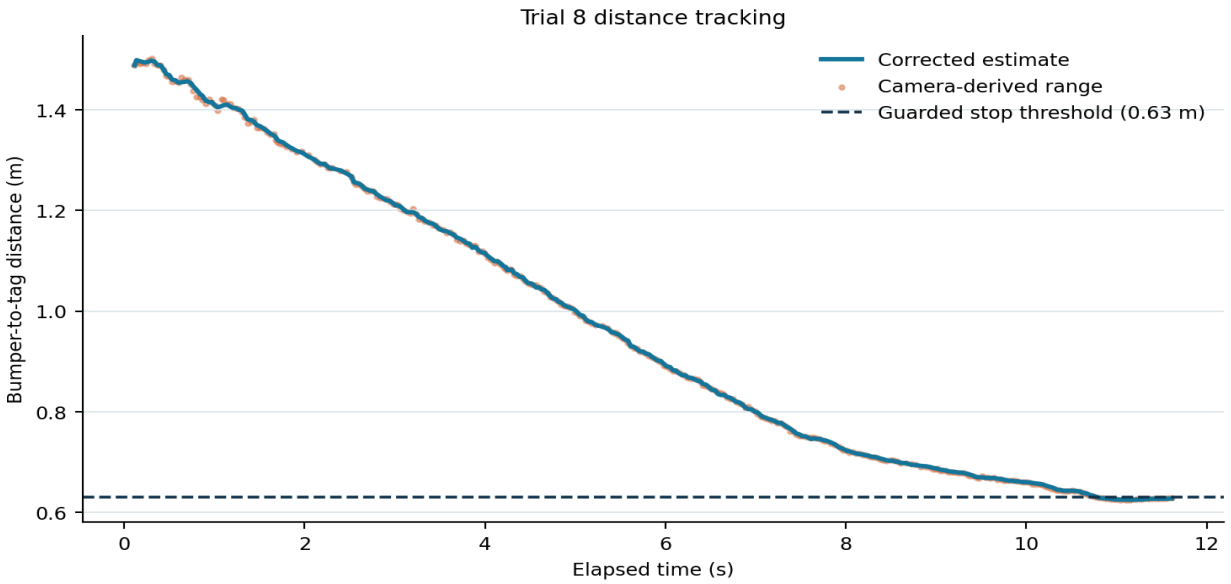


Figure 1. Corrected range and camera-derived range versus elapsed time, with the guarded stop threshold. Source: results/trial_08.csv.

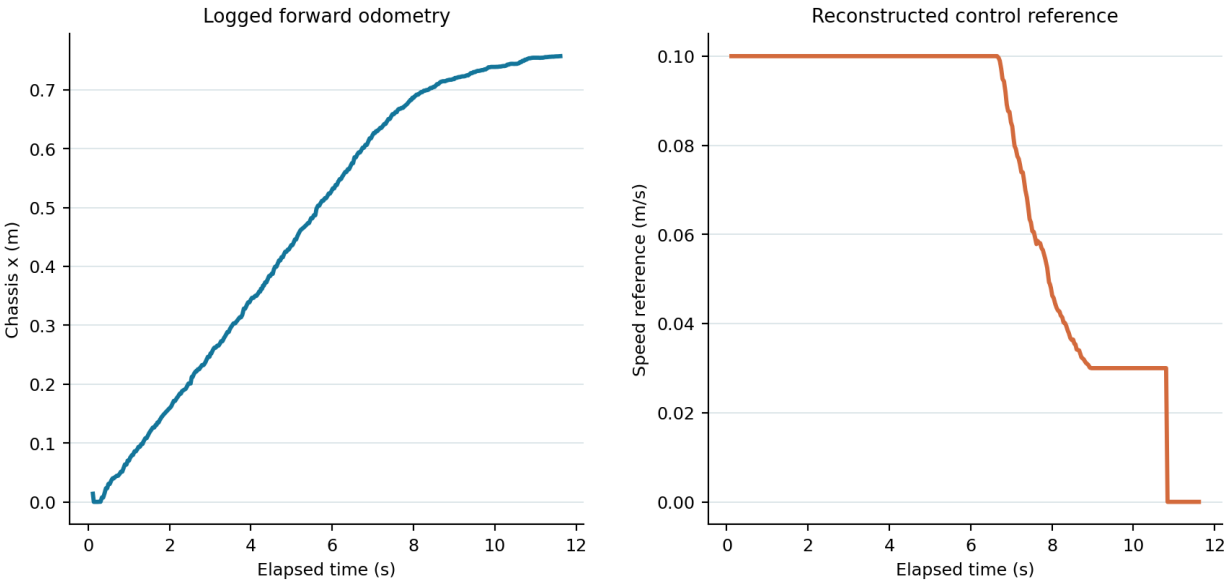


Figure 2. Logged chassis x position and speed reference reconstructed from the logged corrected range using the implemented control law. Individual drive_speed commands and actual wheel speed were not logged.

Correction and calibration

The camera measurement is used only when its innovation relative to the odometry prediction is within 0.15 m. The stationary calibration chart separates the two fit points from two independent tape checks.

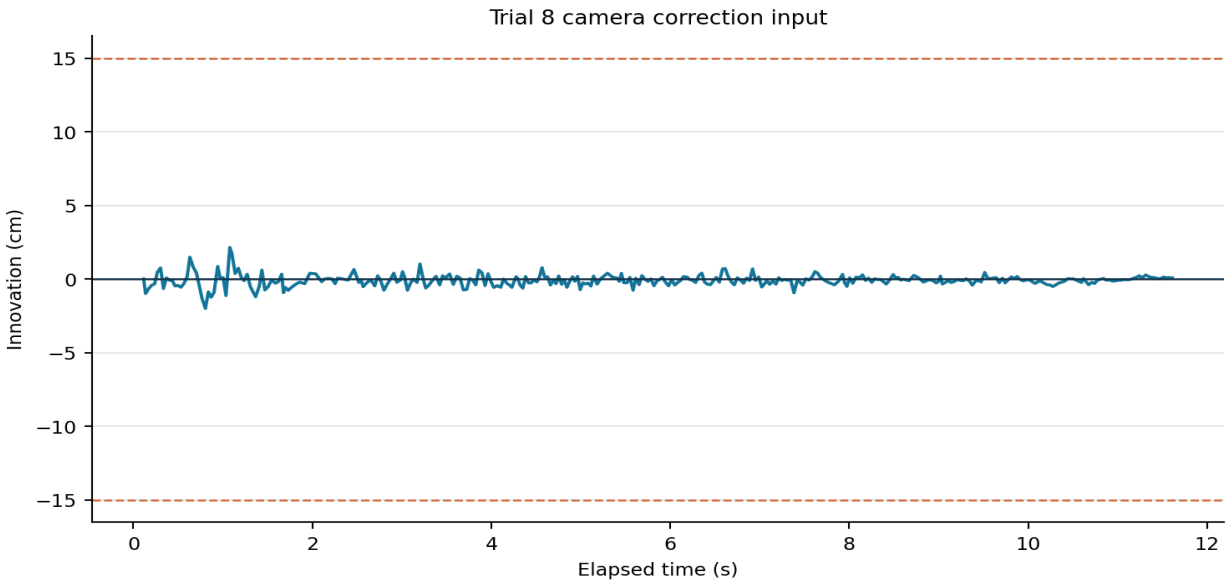


Figure 3. Camera minus predicted bumper range for accepted Trial 8 corrections. Dashed lines show the +/-15 cm acceptance limit. Source: results/trial_08.csv.

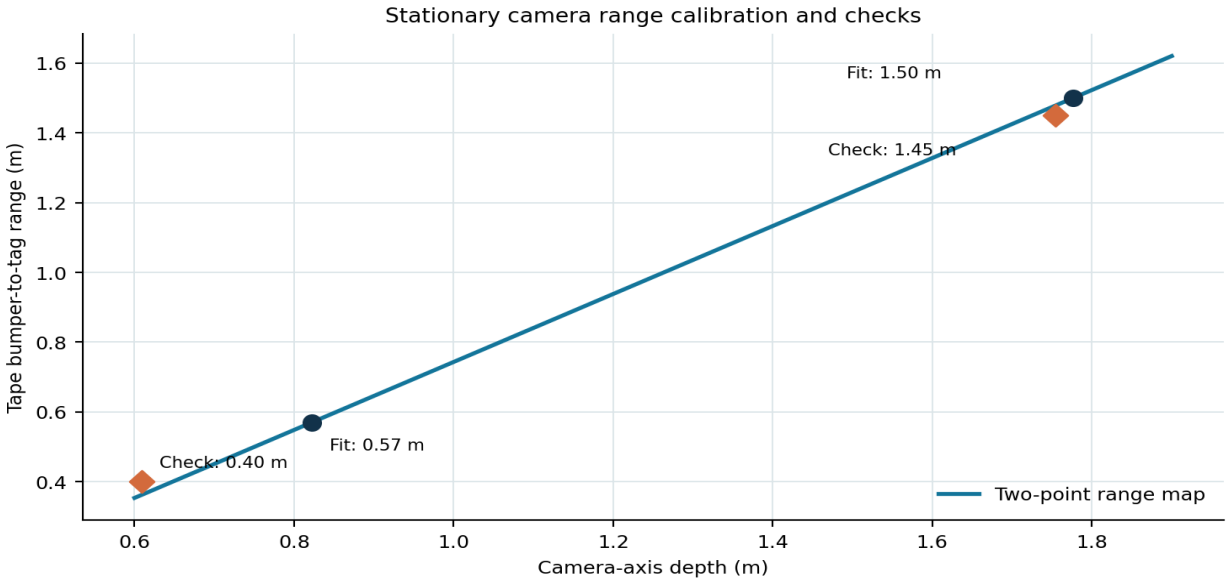


Figure 4. Camera-axis depth versus tape-measured bumper range. The 0.57 m and 1.50 m points define the line; 0.40 m and 1.45 m are held-out checks. Sources: results/preflight_*.json and config/phase3.yaml.

Across-trial comparison

Only eight target-stop runs are included. Trial 7 was an early safety stop and is discussed on page 2. Trial 1's tape value was reported as almost 61 cm, so its plotted measurement and error are approximate.

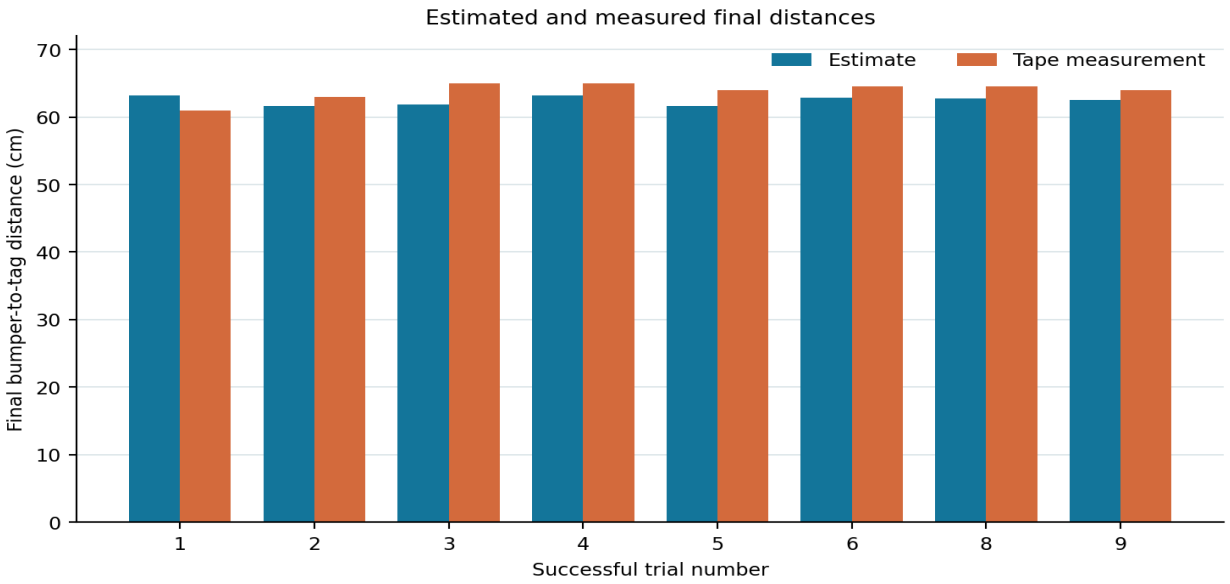


Figure 5. Estimated and independently measured final bumper-to-tag distance for every successful target-stop trial. Source: results/summary.csv.

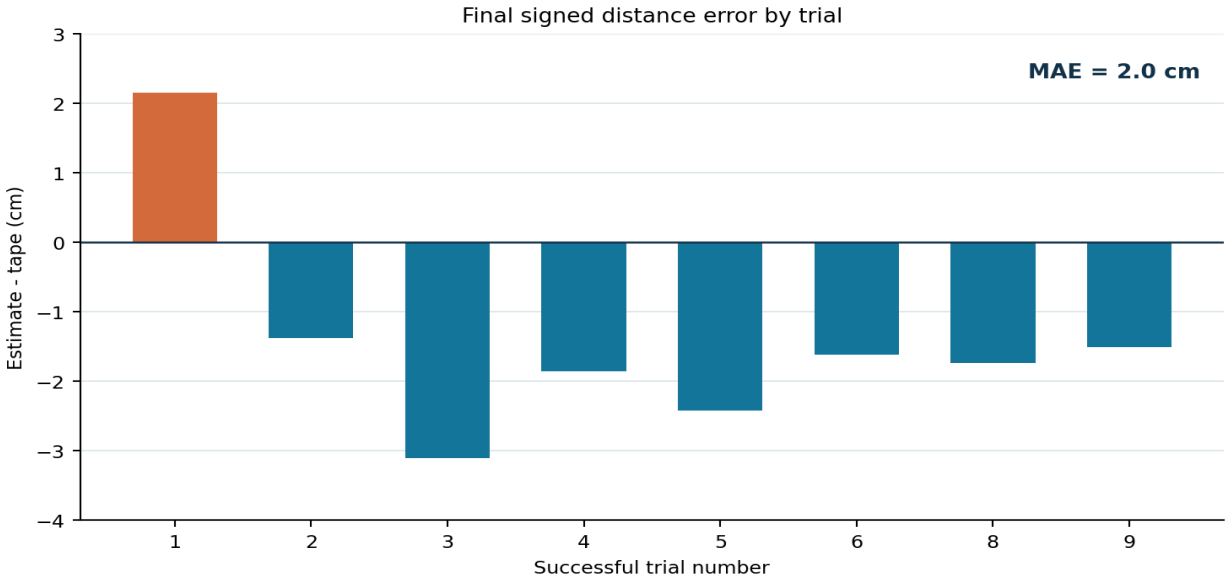


Figure 6. Signed error is estimate minus tape measurement. Mean absolute error (MAE) is 2.0 cm; the signed range is -3.1 to +2.2 cm. Source: results/summary.csv.